

Rayaan Attari

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EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Science <i>M.S. in Computer Science, Machine Learning Track</i> <i>Coursework: Applied ML, NLP, Unsupervised Learning, Deep Learning, Big Data Analytics, Distributed Systems</i>	New York, NY GPA 3.78 Dec 2025
Grinnell College <i>B.A. in Computer Science and Mathematics</i>	Grinnell, IA GPA 3.94 May 2022

SKILLS

Languages & Backend: Python, SQL, TypeScript, Java, FastAPI, Docker, Kubernetes
AI Orchestration: LangGraph, Llamaindex, Claude/Gemini API, RAG, LoRA/QLoRA Fine-Tuning, Prompt Engineering
Machine Learning: PyTorch, HuggingFace (PEFT/Transformers), Scikit-learn, Statistical Modeling
Data Infrastructure: GCP (Certified), AWS, PostgreSQL, Vector Databases (Pinecone, FAISS)

PROFESSIONAL EXPERIENCE

Routerra Health - Columbia BuildLab <i>Founding AI Engineer</i>	New York, NY Aug 2025 – Jan 2026
- Enabled hospital-grade care at home by accelerating scheduling and logistics 2.5x through combinatorial optimization and predictive modeling in the Clinical Route Optimization and Scheduling (CROS) platform, ensuring faster time-to-care - Built probabilistic time-series models (XGBoost, PyTorch) to generate 95% prediction intervals for intraday patient demand and clinician availability, enabling uncertainty-aware staffing decisions and reducing scheduling overhead by 40% - Architected multi-agent orchestration loops using LangGraph to automate clinical operational reviews, embedding strict HIPAA/PHI data compliance protocols and validation guardrails directly into the agentic workflow - Designed and deployed production ML observability and automated evaluation frameworks to track decision quality, implementing A/B testing and statistical monitoring that boosted clinician utilization by 28% across NYC pilots	
Avati Consulting Solutions <i>Machine Learning Engineer, Risk Systems (Intern)</i>	Mumbai, IN May 2025 – Aug 2025
- Engineered an AI-driven Early Warning System within a risk management and compliance software suite to forecast credit default and portfolio risk 3-6 months in advance using financial, behavioral, and macroeconomic signals across 15+ banks - Architected and deployed predictive risk models using statistical and ensemble methods (logistic regression, random forests, gradient boosting), achieving 0.78 AUC and enabling proactive detection of at-risk accounts for credit monitoring - Optimized inference with feature pruning and batch scoring, reducing end-to-end latency by 4x for real-time risk evaluation - Collaborated with business stakeholders to translate model outputs into actionable risk insights by building an LLM-based reporting pipeline using automated data labeling and QLoRA fine-tuning, reducing manual report generation by 60%	
Perficient <i>Software Engineer, Cloud Infra & ML</i>	Dallas, TX Jun 2022 – Mar 2024
- Pioneered the end-to-end modernization of large-scale customer analytics platform by architecting an event-driven microservices solution on Google Cloud that increased modularity and reduced average deployment time by 40% - Migrated large-scale on-prem databases to Cloud SQL, designing ETL workflows to cleanse and transform data; streamlined data retrieval and reduced query response time by 3x, improving SLA compliance for customer-facing applications - Engineered and deployed high-performance, low-latency RESTful APIs and ML inference pipelines for real-time predictive analytics using Vertex AI; integrated Cloud Firestore and Pub/Sub to achieve sub-50ms data processing times for queries - Implemented automated model training, evaluation, and deployment pipelines with Bayesian hyperparameter optimization for sales reward models, reducing retraining time by 30% while serving 100k+ concurrent users	

PROJECTS & RESEARCH

LifeOS: Personal Planning & Decision Agent <i>Python, Claude, LangGraph, React, Vite</i>	Feb 2026
Architected a multi-agent orchestration system to decompose natural language into stateful, cyclic task graphs; scaled to 400+ users by implementing persistent state recovery and automated validation loops to handle LLM non-determinism	
Media Watcher: AI-powered News Intelligence <i>Python, Gemini, RAG, Sentiment Analysis</i>	Jan 2026
Engineered a RAG-based intelligence pipeline utilizing search grounding to extract verifiable risk scores from 2.5k+ news sources; developed an automated evaluation framework that reduced manual compliance review time by 70%	
Spike Sorting: Efficient Dimensionality Reduction for Neural Recordings <i>Python, PyTorch, Scikit-Learn</i>	Dec 2025
Benchmarked 20+ dimensionality reduction and clustering methods (UMAP, t-SNE, VAEs) on large-scale neural embeddings, achieving 10x compression with 85% information retention; results under review for publication	